



Measuring How NBA Players Help Their Teams Win

By [Dan T. Rosenbaum](#)

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I. Introduction

"Good players lead their teams to *wins*." "Lots of players can fill up a stat sheet, but only the great ones are *difference-makers*." "The objective of a basketball game is not to accumulate points or rebounds or assists, *but to win*. What statistic do you have for that?" When I talk with knowledgeable basketball people who are skeptical of statistical analysis, I hear variants on these statements over and over again. The argument is that winning is important and game statistics are only an imperfect measure of many of the contributions that players make to winning.

I very much take this argument to heart. Basketball is not like baseball, a game structured around repeated one-on-one contests between pitchers and batters, where the contributions to winning of any given player can be measured well by individual game statistics. Basketball is much more of a team game, and as noted by [Dean Oliver, one of the leaders of statistical analysis in basketball](#), "[teamwork is the element of basketball most difficult to capture in any quantitative sense](#)" (p. 77). While this argument often is overstated (as much of this teamwork *can* be measured using game statistics), the point still stands. These limitations have led to new approaches to measuring the value of basketball players, approaches that make little use of game statistics like points, rebounds, and assists.

The most common approach is to compute plus/minus ratings that measure how point differentials change when a particular player is in the game versus when he is not. Hockey has used such a plus/minus system for years, and now 82games.com is the first to make these data available for the NBA. The logic of this approach is straightforward; teams should perform better when their good players are playing versus when they are not. The intuitive appeal of this approach has not escaped teams' attention, and my understanding is that most teams use plus/minus ratings to some extent. However, these "unadjusted" plus/minus ratings do not measure the value of a player *per se*; they measure the value of the player relative to the players that substitute in for him. In addition, there are differences in the quality of players that players play with and against. A weak starter on a team with exceptionally good starters (relative to bench players) will generally get a very good unadjusted plus/minus rating – regardless of their actual contribution to the team.

Thus, a better measure of player value would "adjust" these plus/minus ratings to account for the quality of players that a given player plays with and against. In addition, it would account for home court advantage and for clutch time/garbage time play. Thus, unlike in unadjusted plus/minus ratings, these "adjusted" plus/minus ratings do not reward players simply for being fortunate to being playing with teammates better than their opponents. Contributions for individual players are isolated statistically. In this article, I develop adjusted plus/minus ratings [similar to the WINVAL ratings designed by Jeff Sagarin and Wayne Winston](#).^[1] I improve on past efforts by combining estimates of player value using both pure adjusted plus/minus ratings and a statistical index derived from these pure adjusted plus/minus ratings. This hybrid approach leads to player ratings that [unlike press accounts of WINVAL ratings, pass the "laugh test"](#) (p. 181). In addition, the results from this approach are even less noisy than ratings based on traditional statistical indices alone.

Using data from the 2002-03 and 2003-04 seasons (with the latter season being weighted twice as heavily), I find that Kevin Garnett, Tracy McGrady, Andrei Kirilenko, Tim Duncan, and Shaquille O'Neal are the five most effective players in the NBA. Replacing an average player with one of these five players would result in a team improving by about 14 points per 100 possessions or a little over 10 points per game. In other words, in 2003-04 replacing one of the average players on the Orlando Magic with one of these five players likely would have made them a bit better than the New Jersey Nets and Memphis Grizzlies.

Perhaps more importantly, with these adjusted plus/minus ratings I am able to estimate what game statistics predict better performance on the court; these results help explain why certain players have such high adjusted plus/minus ratings. It appears that rebounds are less valuable than typically assumed and steals, blocks, and avoiding turnovers are more valuable. It also appears that having three point shooters on the floor helps teams and that players that can do it all – score, rebound, and assist – are more valuable than simply the sum of those game statistics. In addition, even after accounting for all of those game statistics, players who play more minutes tend to be more valuable for their teams. This finding suggests that coaches recognize those contributions to the team that are not measured by game statistics, and they play those players more minutes.

In this document I lay out a lot of the details of what I am doing and many of you may want to skip over those details. If you just want to see my bottom-line ratings, go to [Table 4](#) or [Table 5](#).

II. A Discussion of the Set-up and Results

Here is the set-up that I use. Every observation is a unit of time in a game where no substitutions are made. There are more than 60,000 such observations per year in 2002-03 and 2003-04. With these data I run the following regression.

$$(1) \quad \text{MARGIN} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K + \varepsilon, \text{ where}$$

$$\text{MARGIN} = 100 * (\text{home team points per possession} - \text{away team points per possession})[2]$$

$X_1 = 1$ if player 1 is playing at home, $= -1$ if player 1 is playing away, $= 0$ if player 1 is not playing

$X_K = 1$ if player K is playing at home, $= -1$ if player K is playing away, $= 0$ if player K is not playing

$\varepsilon = \text{i.i.d. error term}$

β_0 measures the average home court advantage across all teams

β_1 measures the difference between player 1 and the reference players, holding the other players constant

β_K measures the difference between player K and the reference players, holding the other players constant

The reference players are all players playing less than 250 minutes in both seasons combined. Observations are weighted by the number of possessions with (1) observations in 2003-04 weighted twice as heavily as those in 2002-03 and (2) higher weights during crunch time and lesser (or zero) weights during garbage time.[3]

It is in this regression where the effects of the other players on the floor are accounted for. The β s in equation (1) measure the point differential difference (measured per 100 possessions) of the given player relative to the reference players, holding constant all of the players that shared the floor with that player (and with the reference players), i.e. holding the other players constant. What does this “holding other players constant” mean? Strictly speaking, it means that we can take a player and surround him with four teammates and five opponents and compare how that player’s team would do versus how it would do if he was replaced by a replacement player keeping all of the other players the same. This is what is meant by “holding the other players constant,” since we can repeat this exercise with any other combination of other players.

Another way to think of these β s is that they are plus/minus statistics adjusted for the other players on the floor. This takes out the effect of a player who is fortunate to always play with Kevin Garnett or unfortunate enough to always being matched with rookies or NBDL players.

Table 1 presents the results from equation (1) for the top twenty players among those playing 250 minutes or more in the 2002-03 and 2003-04 seasons combined. (I normalize these ratings so that the average player is given a value of zero.) Kevin Garnett has by far and away the highest “pure” adjusted plus/minus statistic, being a full 19.3 points per 100 possessions better than the average player. In addition, this estimate for Garnett is quite precise with it being statistically significantly different from most the rest of the players in the top ten. The rest of the top ten (with the exception of Nenê and players with high standard errors and less than 1,000 minutes) are among the top players in the game – Vince Carter, Andrei Kirilenko, Dirk Nowitski, Tim Duncan, and Shaquille O’Neal. The next ten contains another five players among the top players in the NBA – Rasheed Wallace, Ray Allen, Tracy McGrady, Baron Davis, and John Stockton.

That said, there are a number of outliers in the top 20. Six of those nine outliers (Richie Frahm, Jason Hart, Mike Sweetney, Mickael Pietrus, Earl Watson, and Carlos Arroyo) have standard errors ranging from 4.3 to 6.3, so these players’ high ratings could mostly reflect sampling variation – although Frahm’s rating is so high that despite the high standard error and low number of minutes, it probably is something more than sampling variation. Three players (Nenê, Jeff Foster, and Eric Williams) seem to have genuinely quite good ratings that cannot be explained away by sampling variation. Foster replaced an All-Star in Brad Miller and his team did not miss a beat, ending up with the best record in the League. Nenê played major minutes for a team that improved dramatically in 2003-04, and Eric Williams played on two teams (Boston and Cleveland) that played their best basketball of the season while he was with them.

However, even taking all of that into account, these ratings are quite noisy. Another approach is probably necessary for these rating to be that useful. Below I outline that approach which combines these pure adjusted plus/minus ratings with ratings derived from the relationship between game statistics and these pure adjusted plus/minus ratings.

(I also present offensive and defensive ratings that are based on the pure adjusted plus/minus rating plus an “efficiency” rating that measures how many points per possession are scored by both teams when a given player is one the floor. By combining these two measures, I create offensive and defensive ratings. However, given that I am using two imprecisely estimated ratings to arrive at these offensive ratings, I suspect these rating are measured with quite a bit of error.)

Table 1: Pure Adjusted Plus/Minus Ratings for the Top 20 Players in 2002-03 and 2003-04

[\(Full Table for All Players\)](#)

Rank	Name		Pure Adj. +/-		Offensive		Defensive		Poss. Used	Offensive Efficiency	Total Minutes
	First	Last	Rating	SE	Rating	Rank	Rating	Rank			

1	Kevin	Garnett	19.3	3.0	113.7	2	94.4	15	28%	108	6,553
2	Richie	Frahm	17.3	6.3	114.0	1	96.7	54	15%	126	466
3	Nenê		11.9	2.7	104.3	43	92.4	5	18%	101	4,755
4	Vince	Carter	11.1	2.5	108.1	9	97.0	69	30%	101	4,255
5	Andrei	Kirilenko	11.1	2.6	108.6	8	97.5	89	22%	106	5,108
6	Dirk	Nowitzki	10.6	2.7	109.8	5	99.2	176	24%	115	6,033
7	Tim	Duncan	10.3	3.3	107.2	14	96.8	59	28%	106	5,705
8	Jason	Hart	10.1	5.6	100.7	139	90.6	2	15%	99	660
9	Mike	Sweetney	10.0	5.8	106.9	17	97.0	66	19%	104	495
10	Shaquille	O'Neal	9.9	3.0	107.8	12	97.9	108	27%	110	4,999
11	Rasheed	Wallace	9.6	2.1	105.3	27	95.7	31	22%	105	5,073
12	Mickael	Pietrus	9.5	4.3	106.8	19	97.3	81	18%	102	748
13	Ray	Allen	9.0	2.3	109.7	6	100.7	248	28%	110	5,029
14	Tracy	McGrady	8.6	2.7	109.9	4	101.3	277	32%	111	5,630
15	Earl	Watson	8.1	4.5	107.0	15	98.9	159	21%	90	3,036
16	Jeff	Foster	7.7	2.7	103.7	52	95.9	37	14%	108	2,774
17	Baron	Davis	7.6	2.5	104.4	40	96.8	58	29%	100	4,575
18	John	Stockton	7.3	5.3	104.8	32	97.5	86	23%	106	2,275
19	Eric	Williams	7.1	2.1	104.4	39	97.3	82	18%	102	4,237
20	Carlos	Arroyo	7.1	5.3	112.7	3	105.6	404	25%	101	2,293

Notes: This table is limited to the top 20 of the 420 players playing 250 minutes or more in 2002-03 and 2003-04 combined plus an aggregated observation for those playing less than 250 minutes. Observations from 2003-04 are weighted twice as heavily as those in 2002-03 and clutch time is given a higher weight and garbage time is given a lesser (or zero) weight. The *Pure Adj. +/- Rating* gives the coefficient estimates from regression (1) above normalized so that the average player is equal to zero and is measured in points per 100 possessions. The *SE* column gives the standard error for that rating. *Offensive* and *Defensive Ratings* are measured in points per 100 possessions and I also give the *Rank* out of the 420 players. *Poss. Used* give the percentage of possessions used by a player with field goal attempts, free throw attempts, assists, turnovers, and offensive rebounds with the average being one fifth or 20%. *Offensive Efficiency* gives the average points scored per 100 possessions used, while *Total Minutes* gives the minutes played in 2002-03 and 2003-04.

One approach to dealing with the imprecision of the pure adjusted plus/minus ratings is to use game statistics (points, rebounds, assists, etc.) to measure the correlation between these game statistics and the pure adjusted plus/minus rating. This indirectly measures how these game statistics correlate with point differentials. Equation (2) shows how this can be done in a regression framework.

$$(2) \quad Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_{13} X_{14} + \varepsilon, \text{ where}$$

Y is the pure adjusted plus minus statistic (the β s from the first regression)

X_1 through X_{14} are game statistics described below in Table 2

Table 2 presents the Ordinary Least Squares (OLS) estimates and standard errors from equation (2), along with sample means and standard deviations for each of the variables. The effects of most of the game statistics are as expected with points, rebounds, assists, steals, and blocks having a positive effect and field goal attempts, turnovers, and personal fouls having a negative effect.

Because of the inclusion of the versatility variables multiplying points, rebounds, assists, the marginal effect of another point (or rebound or assist) is not simply given by its coefficient estimate. Evaluated at the mean level of points, rebounds, and assists, the marginal effect of an additional point, assist, offensive rebound, and defensive rebound is 1.08, 1.14, 0.78, and 0.11 points per 100 possessions, respectively. But because the versatility coefficient is positive, these marginal effects are much higher for players with lots of points, rebounds, and assists. Also, note that once we account for the versatility of players, offensive rebounders are more valuable than defensive rebounders (although this difference is not statistically significant). My earlier results showing offensive rebounders being of less value probably was due to negative effect of having offensive rebounding specialists on the floor who don't score or assist much. This negative effect is already accounted for in the versatility measure.

At the mean level of effective field goal attempts, the marginal cost of another two point field goal attempt is equal to 1.09 points per 100 possessions, but at 25 effective field goal attempts per 40 minutes, the marginal cost is just 0.58 points per 100 possessions. This declining marginal cost likely reflects the value of players who can generate field goal attempts as the shot clock is expiring or under extreme defensive pressure. Players who attempt lots of three points and free throws appear to be more valuable than players who specialize in two point field goal attempts. Steals and blocks both appear to be quite valuable, while turnovers are very costly. Personal fouls have a small, but statistically insignificant, *positive* effect, which could account for the possibility that better defenders tend to guard players who generate more fouls.

I also include minutes played per game, and the results suggest that holding all of these other game statistics constant, players who play more minutes tend to help their team point differential. This result would be expected if coaches observe and reward contributions not picked up in game statistics (e.g. good defense) by playing those players more minutes. Note, however, that the coefficient is not huge. Holding the other game statistics constant, the difference between a 20 minutes per game player and a 40 minutes per game player is only 2.16 points per 100 possessions – about the same as an extra steal per 40 minutes.

In the future I hope to add height and age/experience to these regressions. It appears to me that looking at the results in many of the tables that young, inexperienced players tend to have lower pure adjusted plus/minus ratings than their game statistics would suggest. It seems that the young, inexperienced players may not contribute as much to their teams in ways not picked up by game statistics.

Table 2: OLS Estimates of the Effect of Game Statistics on Pure Adjusted Plus/Minus Ratings

Basketball	Coefficient	Standard	Sample	Standard
Statistic	Estimate	Error	Mean	Deviation
Pure adjusted plus/minus rating (dependent variable)	N/A		0.00	4.99
Points per 40 minutes	0.757	0.231	15.55	4.37
Effective field goal attempts per 40 minutes	-1.879	0.402	15.03	3.89
(Effective field goal attempts per 40 minutes) ²	0.026	0.010	241.1	123.3
Three point attempts per 40 minutes	0.323	0.139	2.46	2.21
Free throw attempts per 40 minutes	0.366	0.217	4.02	1.78
Assists per 40 minutes	-0.293	0.277	3.54	2.09
Offensive rebounds per 40 minutes	0.054	0.333	2.00	1.23
Defensive rebounds per 40 minutes	-0.615	0.271	4.99	1.82
Turnovers per 40 minutes	-2.171	0.480	2.36	0.65
Steals per 40 minutes	2.166	0.496	1.31	0.45
Blocks per 40 minutes	1.564	0.327	0.84	0.84
Personal fouls per 40 minutes	0.392	0.281	3.57	1.13
(Points per 40 * Assists per 40 * Rebounds per 40) ^{1/3}	2.262	0.486	6.71	1.38
Minutes per game	0.108	0.037	27.48	8.44
Intercept	-0.043	3.083	N/A	
R ²	0.4397			
Sample size	420			

Notes: This table is limited to the top 20 of the 420 players playing 250 minutes or more in 2002-03 and 2003-04 combined plus an aggregated observation for those playing less than 250 minutes. All of the game statistics are measured per 40 minutes and are pace-adjusted in order to account for teams who average more possessions per minute played. The regression is weighted by minutes played with the 2003-04 season counting twice as much as the 2002-03 season. *Effective field goal attempts* are equal to field goal attempts plus 0.44 times free throw attempts.

With the coefficient estimates in Table 2, it is possible to estimate statistical plus/minus ratings, which is what I do in Table 3. These ratings are much cleaner than those in Table 1, which is evident in two ways. First, the standard errors are generally much smaller in Table 3.^[4] The second is that there are fewer odd players in this list, which is a function of the smaller standard errors, along with expectations that are derived from seeing similar games statistics-based ratings in the past. Note, however, that six of the top 10 using statistical plus/minus ratings in Table 3 are in the top 20 on the pure adjusted plus/minus ratings in Table 1. And arguably, only Brian Cardinal and Shawn Bradley are true outliers in these ratings.

Table 3: Statistical Plus/Minus Ratings for the Top 20 Players in 2002-03 and 2003-04

[\(Full Table for All Players\)](#)

Rank	Name		Statistical		Poss. Used	Offensive Efficiency	Total Minutes
	First	Last	Rating	SE			
1	Tracy	McGrady	15.8	2.6	32%	111	5,630
2	Kevin	Garnett	15.3	2.5	28%	108	6,553
3	Andrei	Kirilenko	13.2	2.5	22%	106	5,108
4	Kobe	Bryant	12.6	2.4	30%	108	5,849
5	Shaquille	O'Neal	12.4	2.6	27%	110	4,999
6	Tim	Duncan	12.3	2.5	28%	106	5,705
7	Dirk	Nowitzki	9.5	2.5	24%	115	6,033
8	Elton	Brand	9.1	2.4	23%	108	5,129
9	Arvydas	Sabonis	8.7	3.8	21%	105	1,209
10	Brad	Miller	8.0	2.3	19%	109	4,893
11	Ray	Allen	8.0	2.4	28%	110	5,029
12	Baron	Davis	7.7	2.5	29%	100	4,575
13	Brent	Barry	7.4	2.4	18%	110	4,285
14	Brian	Cardinal	7.1	2.3	19%	118	1,648
15	Paul	Pierce	7.1	2.4	30%	100	6,195
16	Shawn	Bradley	6.9	2.7	14%	111	2,504
17	Jason	Kidd	6.8	2.4	28%	97	5,438
18	Shawn	Marion	6.7	2.2	22%	106	6,591
19	Ron	Artest	6.6	2.3	25%	100	5,030
20	Sam	Cassell	6.5	2.3	28%	110	5,538

Notes: This table is limited to the top 20 of the 420 players playing 250 minutes or more in 2002-03 and 2003-04 combined plus an aggregated observation for those playing less than 250 minutes. The *Statistical Rating* is estimated using equation (2) and is measured in points per 100 possessions. The *SE* column gives the standard error for that rating. *Poss. Used* give the percentage of possessions used by a player with field goal attempts, free throw attempts, assists, turnovers, and offensive rebounds with the average being one fifth or 20%. *Offensive Efficiency* gives the average points scored per 100 possessions used, while *Total Minutes* gives the minutes played in 2002-03 and 2003-04.

In Table 4 I present “overall” plus/minus ratings that combine the pure adjusted plus/minus ratings in Table 1 and statistical plus/minus ratings in Table 3. The equation for doing so is the following.

$$(3) \quad \text{OVERALL} = \alpha * \text{PURE} + (1 - \alpha) * \text{STATS}, \text{ where}$$

OVERALL is the overall plus/minus rating

PURE is the pure adjusted plus/minus rating from Table 1

STATS is the statistical plus/minus rating from Table 3

α is the share of the overall rating due to the pure rating (it is chosen to minimize the standard error of the overall rating with the restriction that it fall between 10% and 90%, note that this will result in the pure rating counting less when it is especially noisy)

These overall plus/minus ratings presented in Table 4 containing very few outliers. Brian Cardinal is probably the only true outlier, but over the past two seasons he has been remarkably efficient with the highest offensive efficiency (tied with Peja Stojakovic) among the top 20 players. These results confirm that the players recognized among the best in the NBA also are the most instrumental in improving their teams’ point differentials. The surprises in this list are that (1) Kevin Garnett is head and shoulders above any other player, (2) Andrei Kirilenko is an elite player in the NBA, and (3) Vince Carter and Rasheed Wallace are much more effective players than is commonly assumed.

Table 4: Overall Plus/Minus Ratings for the Top 20 Players in 2002-03 and 2003-04

[\(Full Table for All Players\)](#)

Rank	Name		Overall		α	Pure Adj. +/-		Statistical		Poss. Used	Offensive Efficiency	Total Minutes
	First	Last	Rating	SE		Rating	SE	Rating	SE			
1	Kevin	Garnett	16.2	2.4	23%	19.3	3.0	15.3	2.5	28%	108	6,553
2	Tracy	McGrady	12.6	2.4	44%	8.6	2.7	15.8	2.6	32%	111	5,630
3	Andrei	Kirilenko	12.3	2.4	42%	11.1	2.6	13.2	2.5	22%	106	5,108
4	Tim	Duncan	12.1	2.5	11%	10.3	3.3	12.3	2.5	28%	106	5,705
5	Shaquille	O’Neal	11.8	2.5	26%	9.9	3.0	12.4	2.6	27%	110	4,999
6	Kobe	Bryant	10.0	2.4	31%	4.3	2.8	12.6	2.4	30%	108	5,849
7	Dirk	Nowitzki	9.9	2.4	34%	10.6	2.7	9.5	2.5	24%	115	6,033
8	Ray	Allen	8.5	2.1	53%	9.0	2.3	8.0	2.4	28%	110	5,029
9	Baron	Davis	7.6	2.3	53%	7.6	2.5	7.7	2.5	29%	100	4,575
10	Vince	Carter	7.6	2.2	37%	11.1	2.5	5.6	2.3	30%	101	4,255
11	Elton	Brand	7.6	2.3	20%	1.5	2.9	9.1	2.4	23%	108	5,129
12	Arvydas	Sabonis	7.4	3.7	17%	1.1	4.7	8.7	3.8	21%	105	1,209
13	Rasheed	Wallace	7.4	2.0	61%	9.6	2.1	4.0	2.3	22%	105	5,073
14	Jason	Kidd	6.9	2.5	10%	7.0	3.7	6.8	2.4	28%	97	5,438
15	Brian	Cardinal	6.8	2.3	13%	4.8	3.0	7.1	2.3	19%	118	1,648
16	Paul	Pierce	6.8	2.2	36%	6.2	2.6	7.1	2.4	30%	100	6,195
17	Shawn	Marion	6.6	2.2	15%	5.8	2.8	6.7	2.2	22%	106	6,591
18	John	Stockton	6.4	3.1	10%	7.3	5.3	6.3	3.1	23%	106	2,275
19	Ron	Artest	6.3	2.2	44%	6.0	2.4	6.6	2.3	25%	100	5,030
20	Predrag	Stojakovic	5.4	2.3	10%	2.1	3.3	5.8	2.3	21%	118	5,712

Notes: This table is limited to the top 20 of the 420 players playing 250 minutes or more in 2002-03 and 2003-04 combined plus an aggregated observation for those playing less than 250 minutes. The *Overall Rating* combines the pure adjusted plus/minus ratings and statistical plus/minus ratings as described in equation (3) and is measured in points per 100 possessions. The *SE* column gives the standard error for that rating. α is the share of the *Overall Rating* due to the pure adjusted plus/minus rating. The *Pure Adj. +/- Rating* gives the coefficient estimates from regression (1) normalized so that the average player is equal to zero and is measured in points per 100 possessions. The *SE* column gives the standard error for that rating. The *Statistical Rating* is estimated using equation (2) and is measured in points per 100 possessions. The *SE* column gives the standard error for that rating. *Poss. Used* give the percentage of possessions used by a player with field goal attempts, free throw attempts, assists, turnovers, and offensive rebounds with the average being one fifth or 20%. *Offensive Efficiency* gives the average points scored per 100 possessions used, while *Total Minutes* gives the minutes played in 2002-03 and 2003-04.

Table 5 is a continuation of Table 4 showing how players in the top 20 of the overall plus/minus ratings rate on various rating systems. It lists the rankings for the pure adjusted plus/minus ratings in Table 1, the statistical plus/minus ratings in Table 3, and the per 40 minutes [efficiency index computed by NBA.com](#). Comparing these rankings shows once again how noisy the pure adjusted plus/minus statistics are.

Table 5: Overall Plus/Minus Ratings and Comparison of Ratings for the Top 20 Players in 2002-03 and 2003-04

[\(Full Table for All Players\)](#)

Rank	Name		Overall		Rankings			Poss. Used	Offensive Efficiency	Total Minutes
	First	Last	Rating	SE	Pure	Stats	Index			
1	Kevin	Garnett	16.2	2.4	1	2	1	28%	108	6,553
2	Tracy	McGrady	12.6	2.4	14	1	7	32%	111	5,630
3	Andrei	Kirilenko	12.3	2.4	5	3	14	22%	106	5,108
4	Tim	Duncan	12.1	2.5	7	6	2	28%	106	5,705

5	Shaquille	O'Neal	11.8	2.5	10	5	3	27%	110	4,999
6	Kobe	Bryant	10.0	2.4	52	4	10	30%	108	5,849
7	Dirk	Nowitzki	9.9	2.4	6	7	4	24%	115	6,033
8	Ray	Allen	8.5	2.1	13	11	31	28%	110	5,029
9	Baron	Davis	7.6	2.3	17	12	78	29%	100	4,575
10	Vince	Carter	7.6	2.2	4	27	51	30%	101	4,255
11	Elton	Brand	7.6	2.3	109	8	8	23%	108	5,129
12	Arvydas	Sabonis	7.4	3.7	122	9	5	21%	105	1,209
13	Rasheed	Wallace	7.4	2.0	11	43	52	22%	105	5,073
14	Jason	Kidd	6.9	2.5	21	17	21	28%	97	5,438
15	Brian	Cardinal	6.8	2.3	44	14	38	19%	118	1,648
16	Paul	Pierce	6.8	2.2	26	15	40	30%	100	6,195
17	Shawn	Marion	6.6	2.2	31	18	30	22%	106	6,591
18	John	Stockton	6.4	3.1	18	21	19	23%	106	2,275
19	Ron	Artest	6.3	2.2	29	19	90	25%	100	5,030
20	Predrag	Stojakovic	5.4	2.3	92	26	36	21%	118	5,712

Notes: This table is limited to the top 20 of the 420 players playing 250 minutes or more in 2002-03 and 2003-04 combined plus an aggregated observation for those playing less than 250 minutes. The *Overall Rating* combines the pure adjusted plus/minus ratings and statistical plus/minus ratings as described in equation (3) and is measured in points per 100 possessions. The *SE* column gives the standard error for that rating. Rankings out of the 420 players are given for the pure adjusted plus/minus ratings in Table 1 (*Pure*), the statistical plus/minus ratings in Table 3 (*Stats*), and the per 40 minutes [efficiency index computed by NBA.com](#) (*Index*). *Poss. Used* give the percentage of possessions used by a player with field goal attempts, free throw attempts, assists, turnovers, and offensive rebounds with the average being one fifth or 20%. *Offensive Efficiency* gives the average points scored per 100 possessions used, while *Total Minutes* gives the minutes played in 2002-03 and 2003-04.

If you have any questions or comments about this analysis, feel free to [call me](#) or [e-mail me](#). When possible, I will incorporate comments on this analysis.

** I thank . . . and other anonymous individuals for comments on this analysis. All errors or mistakes, however, remain my own. Also, all conclusions are my own and do not reflect on anyone who might have aided me in this analysis. This analysis can be used in whole or in part, as long as (1) this piece is credited to Dan Rosenbaum, (2) a link to the latest version of this analysis (<http://www.uncg.edu/eco/rosenbaum/NBA/winval2.htm>) is included, and (3) I have given express permission for this use ([e-mail me](#)).

[1] Jeff Sagarin has successfully applied this methodology to teams for years and to individuals in non-team sports, such as golf and tennis, but this is the first time that I know of that this methodology has been applied to a individuals in a team sport. The Dallas Mavericks reportedly are paying more than \$100,000 per year for this WINVAL system, which according to my knowledge is by far the greatest sum being paid to outside statistical consultants in the NBA.

[2] In an observation where either the home or away team has zero possessions, the average home (or away) points per possession is used instead. If both teams have zero possessions, then the observation is deleted.

[3] Here is my exact code. Clock measures the minutes elapsed in the game at the beginning of the observation. Three minutes left in the game (in regulation or in overtime) is counted as 45. Margin is the absolute value of the difference in scores at the beginning of the observation.

```
ptime=max(0,(clock-36)/12);
marg10=10-ptime*7;
wgt=10*(1+ptime)*max(0,min(1,(1-(margin/marg10-1))));
```

Basically, in the first three quarters, full weight is given to any part of a game where the margin is less than 10 and no weight is given if it is more than 20. Between 10 and 20, the weight is phased from full to zero.

This is basically what happens in the fourth quarter as well, except that I decrease the margin from 10 (20) to 3 (6) from the beginning to the end of the fourth quarter. Also, ceteris paribus, the end of the quarter counts more than the beginning of the quarter.

At the end of all of this, I renormalize the weights so that on average minutes in the fourth quarter count the same as those in the first three quarters.

[4] This standard error consists of two parts. The first is due to sampling variation from only observing players for a limited number of minutes. The second is due to the assumption that statistics ignore 15 percent of the true contributions of players, implying that even if an infinite number of games were observed, there would still be a standard error associated with the statistical plus/minus rating.

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